

FORMATIVE ASSIGNMENT

CareerBridge Africa

- **Description of the concept of their Mission;**

My mission is to solve the unemployment problem, create change, and innovation through software engineering and cybersecurity skills, and give job opportunities to young Africans for them to thrive. Using technology, I can solve unemployment problems by creating innovative platforms that will be used by African youth so that they can realize their potentials

The use of my mission in job creation and the education sector is based on the fact that unemployment rates among graduates keep increasing day to day, especially in African countries. Through the use of technology, there will be an opportunity to solve unemployment problems among graduates and young Africans

- **Background:**

In many countries across Africa, many young people finish their studies with the hope that they will get jobs and build their careers, but they struggle to find those opportunities, including jobs, internships, mentorship, or career guidance, and networking that can help them build their careers, and this unemployment among youth continues to grow, even among those with skills. (MOHAC AFRICA, April 6, 2026)

Many employers and companies also face the challenge of finding those skilled young people because there's something that's missing between education and opportunities. Although there are online platforms for job openings, such as LinkedIn, many African youths still struggle with finding places where they can access credible opportunities that suit their skills and careers.

This proves that there is an urgent need for an online platform to be established that can help solve this challenge by linking the youth with opportunities and career guidance

- **Problem statement:**

Youth unemployment has become a very serious problem for students and graduates in Africa, especially in developing countries like Rwanda. Every year, there is a considerable number of young people who graduate from Universities, but it seems impossible for them to acquire internships, employment, or mentorship related to their skills and career interests. (ACN, 2023)

This problem affects the following parties: students, graduates, potential employers, and the community in general, because many talented young individuals become unemployed despite the fact that they have received degrees or education. This matter has persisted throughout the year due to the following reasons: Limited access to opportunities, lack of career guidance, skills mismatch, lack of connection, and unreliable job information online.

Currently used social networking sites, like LinkedIn, are concentrated on global markets, but they don't consider all challenges related to the local African market, which young Africans encounter in searching for opportunities. Therefore, many students cannot create useful networks or find useful opportunities for themselves.

The proposed CareerBridge Africa project will help solve this issue because the platform is especially created for African students and graduates who will be able to find relevant jobs, mentorship, and companies within their region.

- **Suggested solution**

The proposed solution involves the creation of a platform known as CareerBridge Africa that will assist students and graduates in creating a profile, finding verified internships and job offers, connecting with mentors, and receiving personalized recommendations on the basis of the user's skills and interests.

CareerBridge Africa differs from other platforms in that it is designed for Africans only and will offer mentorship assistance as well as verified jobs. This way, the issue of fake offers can be minimized, and it will reduce the problem of not getting a job because you don't have a connection or experience.

- **Software development model**

The model that is chosen: Agile Software Development Model

The agile model is relevant to my project because the platform will need to be improved

continuously according to feedback received from graduates, students, and employers. Since users' needs might change as time goes by, the Agile model allows the project to be improved continuously and developed step by step.

Steps that will be used in the project:

1. Requirement gathering:

We will be collecting information from graduates, employers, and students to understand what they need and the challenges they're facing, so that we can know well the problem we need to solve and to know what's missing from existing platforms.

2. Planning:

Identify the main features of the app, such as creating profiles, matching opportunities, and mentorship

3. Design:

Create straightforward designs and prototypes of the user interfaces for testing.

4. Development:

Implement the software into parts called sprints, beginning with its most basic functionalities.

5. Testing:

Do frequent testing of the program, involving users, to improve its usability

6. Deployment:

Deploy the functional prototype of the app for use by graduates and students.

7. Maintenance and Improvement:

Continue to improve the system according to the needs of users

- **The hypothesis of their solution**

However, if CareerBridge Africa is launched and becomes accessible to students and graduates in the universities of Rwanda as the starting point, then within six months of the launch of the platform, at least 60 percent of the registered users who regularly use the platform (i.e., log into the website at least four times a month) will have managed to find at least one opportunity in terms of employment, Internship, or career guidance that is relevant to their skills and career aspirations. The lessons learned in this pilot will then help in expanding CareerBridge Africa to other countries across Africa.

This hypothesis is SMART since:

Specific: The target population consists of the registered Users of the platform in ALU and partner universities in Rwanda who use it.

Measurable: The level of success will be the percentage of active users who find at least one opportunity, measured through surveys and application data in the platform.

Achievable: The platform tackles all the barriers (i.e., lack of verification, career guidance, personalisation) preventing young Africans from getting access to opportunities.

Relevant: This hypothesis relates directly to the key challenge of youth unemployment among African graduates.

Time-bound: The period of evaluation is six months after the launch of the platform, which makes it easy to access once it is deployed.

- **References**

1. **EMPLOYABILITY IN AFRICA Career Opportunities and Job Placement Trends Report.** (ACN, 2023, summary employability report)
<https://careernetworks.africa/wp-content/uploads/2024/04/SUMMARY-Employability-Report-by-ACN-Summary-2023.pdf>
2. **Unemployment Rate in Africa: Causes, Statistics, and Solutions.**
<https://mohacafrica.org/unemployment-rate-in-africa/>

Software Requirements Specification (SRS)

Title: CareerBridge Africa

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29 May 2026

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0.2. Revision History

Name	Date	Reason For Changes	Version
Uwamahoro Shakilla	28th June 2026	Added more functional requirements and improved the hypothesis to SMART format	1.1

1. Introduction

1.1. Purpose

The Software Requirements Specification(SRS) is the details of functional and non-functional requirements for CareerBridge Africa, version 1.0.

CareerBridge Africa is a web app that is designed to solve the issue of unemployment among young Africans, especially in Rwanda. This SRS outlines all aspects of the platform, from user registration to opportunities, mentoring, and personalisation of the service and administration features.

This document aims to create an agreement between developers and other parties concerning the capabilities that the system must have. The SRS will be used for design, development, and testing during the entire process of creating CareerBridge Africa.

1.2. Document Conventions

The following conventions have been followed within this document:

- Bolded text is used to indicate section headings, table headings, and definitions of terms for the first time they appear in the document.
- Function requirement (FR) is denoted by FR X.X while Non-requirement (NFR) is denoted by NFR X.
- All requirements are prioritized equally unless there is a different indication. Prioritization changes would be made through discussion with the facilitator in the course of sprint planning.
- Hierarchical numbering of sections as 1, 1.1, 1.2, and so on facilitates referencing.
- The term ‘User’ can refer to anyone who interacts with the system, e.g., student, graduate, employer, mentor, and administrator, unless specified differently.

1.3. Intended Audience and Reading Suggestions

This software requirements specification is meant for various groups of audience who may benefit from the information contained in it in the following manner:

- Engineers/Developers: Sections 3 & 4 of the specification contain valuable information on external interfaces and functional and non-functional requirements.
- Project managers/Facilitators: An initial look at the product can be obtained by reading sections 1 & 2. Section 5 will give details regarding the quality and performance of the system.
- Testers/QA reviewers: This audience group should focus more on sections 4 & 5, where the system specifications mandatory for acceptance are laid out.
- Readers/Stakeholders, in general, who are interested in knowing the objectives and the background of the system: Those people would have to start with section 1, followed by section 2, before moving to any other section.

Users are requested to begin with section 1 before going to any other subsequent sections of the software requirement specification. Sections of this document may also be read individually using the table of contents provided.

1.4. Product Scope

CareerBridge Africa refers to an online web app that seeks to help close the gap between African youths looking for employment and those employment openings. This is achieved by enabling students and graduates to make personal profiles, browse, and apply for verified Internships and jobs, mentorship and get personalized career advice based on their individual skills.

As far as employers are concerned, they get the chance to post employment positions and find many candidates within Africa. There is also an administrative component to the platform that controls and moderates the activity on it.

Overall, CareerBridge Africa is meant to solve issues such as youth unemployment in developing African nations, including Rwanda. By developing a trusted, Africa-specific website, CareerBridge Africa aims to reduce the information gap between education and employment opportunities.

2. Overall Description

2.1. Product Perspective

CareerBridge Africa is a new, standalone system designed specifically for young Africans. The Platform combines opportunity matching, mentorship, and career development into one system.

Even though there are existing platforms that work like CareerBridge Africa, providing job opportunities, it is mainly for global markets; it doesn't solve the problem of unemployment among African youths, including a lack of verified opportunities, skills mismatch, lack of experience, and career mentorship. The CareerBridge Africa platform is mainly designed for Africa.

2.2. Product Functions

The system will:

- Allow the user to register an account
- Allow to create profiles
- Make recommendations on opportunities
- Allow employers to post job listings
- Provide mentorship tips and tutorials, teaching about how to choose a career that matches skills
- Notify users about new opportunities
- Let users apply for opportunities

2.3. User Classes and Characteristics

1. **Students and Graduates:** They are those who would like to pursue internship positions and job positions, as well as mentorship. They are the key stakeholders when it comes to usage of the app, since although some of them don't have technical knowledge whatsoever, they would use the app in order to search for internships and career guidance.
2. **Employers:** These include Organizations, NGOs, and companies offering Internships and job opportunities.
3. **Admin:** This is the person who has full control of the app; they manage and verify activities on the platform. They are also the ones who will be responsible for providing mentorship tips or career guidance.

2.4. Operating Environment

The platform will operate on:

- Windows
- Android devices
- Web browsers such as Chrome and Edge

2.5. Design and Implementation Constraints

The Design and implementation of CareerBridge Africa will be limited to the following things:

- Internet connection: The app should be used where there is a network connection, and some may have limited internet access because some may be using a public network, and due to many users connected to it, the internet may go slow, and some may not be able to access it.
- Budget: Because the app must be developed within the academic program, the resources and budget are limited.
- Data security: All information about users must be protected. CareerBridge Africa must provide security and data privacy protection according to the laws here in Rwanda, and it requires more processes so that it can be safe from hackers.
- Compatibility with Mobile devices: The platform must be responsive on mobile devices to make sure it will be accessible to everyone.

2.6. User Documentation

The system will provide:

- User guide: a step-by-step guide on how to use the platform, including: how to create an account and log in, how to create a profile, searching and applying for opportunities, and all other things our app must do, and this guide must be available online and in PDF.
- Help pages: The app must have help pages for questions the user might have, and where to submit tickets(the queries). It should include some questions that are commonly asked by users and answers, and those who are not satisfied or have questions that are different might submit a ticket.
- Tutorials for users: Visual guides like videos for first-time users, guiding them on how to use the app, and also for mentoring, which are for those who need career guidance.

2.7. Assumptions and Dependencies

These assumptions have been made during the preparation of this document, so that the app will be developed and deployed well:

- It is assumed that Users have internet access and that it is stable and working well to use the app.
- It is assumed that employers and companies in the African market are willing to post the jobs and internship opportunities.
- It is assumed that the Users have smartphones or computers to access the app.
- It is assumed that Users know how to read, and the platform must support the language that is commonly used, which is English.
- It is assumed that the resources and funds will be available.

3. External Interface Requirements

3.1. User Interfaces

The system will provide simple and user-friendly interfaces for:

- Registration: The registration and login page are among the first things the user will see on screen; they must be clear and straightforward.
- Dashboard: The system will have a dashboard that the user will see next after login, where they will see recommendations on opportunities, recent messages, and any pending notifications.
- Profile creation: The User must be able to create a profile and edit it, to add skills, experience, and e.t.c.

- Opportunity search bar: The search bar must be available on the listings page, allowing users to filter the results by location, category, employment type, and skills.
- All screens must have a navigation bar with quick access to the dashboard, listings, messages or feedback from users, and the user profile.
- Feedback and messaging: Error messages and feedback must be displayed clearly and in an understandable way to users, not in technical language, which can confuse some users who don't have an idea of what they mean.

3.2. Hardware Interfaces

CareerBridge Africa is an online app that doesn't have the need for any special hardware equipment. The software communicates with the standard hardware via the browser on the user's device.

- The application will access the device camera for uploading photographs during profile creation (with the user's permission). The user will be able to take a picture or upload it from the computer, in case they don't have a camera, like when using a desktop.
- The app will make use of the device's GPS to find out locations and offer relevant opportunities (with the user's permission).
- The system will not require any special hardware equipment to function properly, but the main hardware devices that the system will work on are: computers, smartphones, and Tablet

3.3. Software Interfaces

The Platform will interact with the following external software interfaces:

- Database: MySQL or PostgreSQL will be used to store all user data, messages and feedback, listings, and app records.
- Cloud storage: A cloud storage service will be used to store uploaded files of users, such as CVs and profile photos.
- Web browsers: Web browsers will be used to navigate through the platform
- Email notification systems: A third-party service will be used to handle the emails, such as confirmation of registration, password reset, and notifications.

3.4. Communications Interfaces

The system will use the following communication requirements:

- All data that is exchanged between client and server must be protected using HTTPS with TLS 1.2 or higher encryption to protect user information.
- The application will communicate using RESTful APIs between the frontend and backend servers.
- Communication via email will be done over SMTP from a third-party provider service. All notification emails must reach their intended recipients within five minutes of the trigger.
- Messages passed between mentors and mentees will be communicated in near real-time and stored in a database to be pulled on page load or through polling mechanisms.

4. Requirement Specification

STAKEHOLDER REQUIREMENTS SPECIFICATION

4.1. Functional Requirements

Req ID	Requirements	Description
FR 1	User registration & authentication	
FR 1.1	Register account	Allow students, graduates, and employers to create accounts using email or Google Sign-in options
FR 1.2	Login	Authenticate users with secure credentials and allow password reset via email
FR 1.3	Profile setup	Allow users to complete their profile by adding skills, education, and career interests after registration.
FR 1.4	Password reset	Allow users to reset their passwords when they forget their password, there will be “forgot password” option to reset
FR 1.5	Role selection	Allow users to select their role during registration (Student/Graduate or Employer) so the system shows related content
FR 2	Job and internship listing	
FR 2.1	Post opportunities	Allow verified employers to post job opportunities and internships with all requirements.

FR 2.2	Opportunity search	Allow students and graduates to search and filter opportunity listings by field, location, career interests, and experience level
FR 2.3	Apply to opportunities	Allow users to directly apply through the platform by submitting their CV and cover letter
FR 2.4	Save opportunities	Allow users to bookmark listings to review or apply to later.
FR 3	Mentorship Module	
FR 3.1	Tutorials and tips	Allow Admin to provide tutorials and tips for career guidance to those who need them
FR 3.2	Browse guidance content	Allow students and graduates to browse and search all published tips and tutorials by categories of topic
FR 4	Messaging & Notifications	
FR 4.1	Contact admin	Allow Users to send questions and messages to the admin contact form in the platform.
FR 4.2	Opportunity notifications	Allow to notify users about new opportunities via email and in-app alerts
FR 4.3	Application notifications	Notify users when the employer updates the status of their application.
FR 4.4	User feedback	Allow the user to rate and give feedback about the app's usability, or any other feedback
FR 5	Admin Dashboard	
5.1	Employer verification	Employers are required to submit their business registration documents before posting; the admin reviews and approves, or rejects the request
FR 5.2	Admin verification	Allow admin to verify links or opportunities posted by employers, if they are valid and secured, and approve, reject, or remove them
FR 5.3	Manage users and listings	Allow the admin to manage all users, listings, and all other activities made on the platform
FR 5.4	Report fake listings	Allow users to highlight any suspicious listings, and if they are fake, they will be reported
FR 5.5	Platform analytics	Provide the admin with a dashboard showing the total registered users, active listings, applications submitted, and career guidance content reviews

5. Other Nonfunctional Requirements

5.1. Example of Non-Functional Requirements

Requirement Type	Req ID	Description
Security	NFR 1	User Access - Authenticate user on AD
Usability	NFR 2	The language used to communicate with users is English
Cross-Browser Support	NFR 3	The system must be accessible across every web browser.
Technology	NFR 4	The system must be accessible on: • PC • Android On every tool with a browser and display, as well as the internet

5.2. Performance Requirements

CareerBridge Africa needs to offer high and efficient system performance where people logging into the system from different parts of Africa can use the site without experiencing slow loading, which takes seconds only on a good internet.

The system needs to be able to handle several users using the site at the same time efficiently. Notifications like resetting password link, opportunities sent out, and application received notifications need to be done in a few minutes.

5.3. Safety Requirements

This system will protect users from dangerous and fake opportunities through a process where the administrator will first verify the jobs and internship listings before it's posted on the website.

The user must also be able to notify the system of any suspicious links, any harassment, and improper use of the account. User details like emails, resumes, and contact information must be kept well in the system so that they can't fall into the wrong hands.

The system should also be able to do backups of important data to avoid losing it.

5.4. Security Requirements

The platform needs to have strong security protocols to make sure that all user accounts and personal information are protected. Users should log in to the app through encrypted means.

HTTPS encryption must be used to make sure that data sent and received from the users is secured. User credentials, like passwords, must be kept safe in the database. The system must also offer protection from cyber attacks.

5.5. Software Quality Attributes

Usability: The interface should be easy to use, even for those people who don't know tech terms.

Reliability: it should keep running well without failing many times.

Scalability: CareerBridge Africa must be scalable and manage the growing number of users as more young Africans join the platform.

Maintainability: The system should allow further changes without much effort from the developer.

Portability: The app must work on all devices and browsers.

Performance: It should perform fast, even when it is used by many users simultaneously.

5.6. Business Rules

- Only honest employers and organisations will be allowed to post jobs or internships on this app
- A user needs to register an account before applying for opportunities.
- Opportunities given by users will first be verified before being posted.
- Users need to provide correct information when creating a profile and making an application. And any user who is found sharing fake information shall be removed from the app.

6. Appendix

This appendix part contains extra data on CareerBridge Africa to facilitate its understanding and implementation, and the hard words or tech terms used in the SRS.

The system might have the following things in the future: mobile app functionality and support for different African languages.

Appendix A: Glossary

This part includes the terms used in the SRS, the abbreviations, and their definition:

1. **SRS(Software requirements specification):** The document that describes functional and non-functional requirements of a software system.
2. **User:** anyone who will be using the app, including students, employers, and graduates.
3. **Admin:** the person who manages or controls everything that happens in the platform.
4. **Tech terms(Technical terms):** the words mainly used in technology.
5. **Authentication:** The way of checking the identity of a user when he or she is logging in.
6. **API:** Are the rules that allow different software systems to communicate with each other.
7. **HTTPS/TLS:** Are protocols that are basically used to protect data that is shared over the internet.
8. **Agile:** This is the software development technique used by developers, which allows changes and breaks down a difficult project into small parts called sprints.
9. **Database:** This is used to store data from the platform.
10. **CareerBridge Africa:** is the title or the name of the platform.

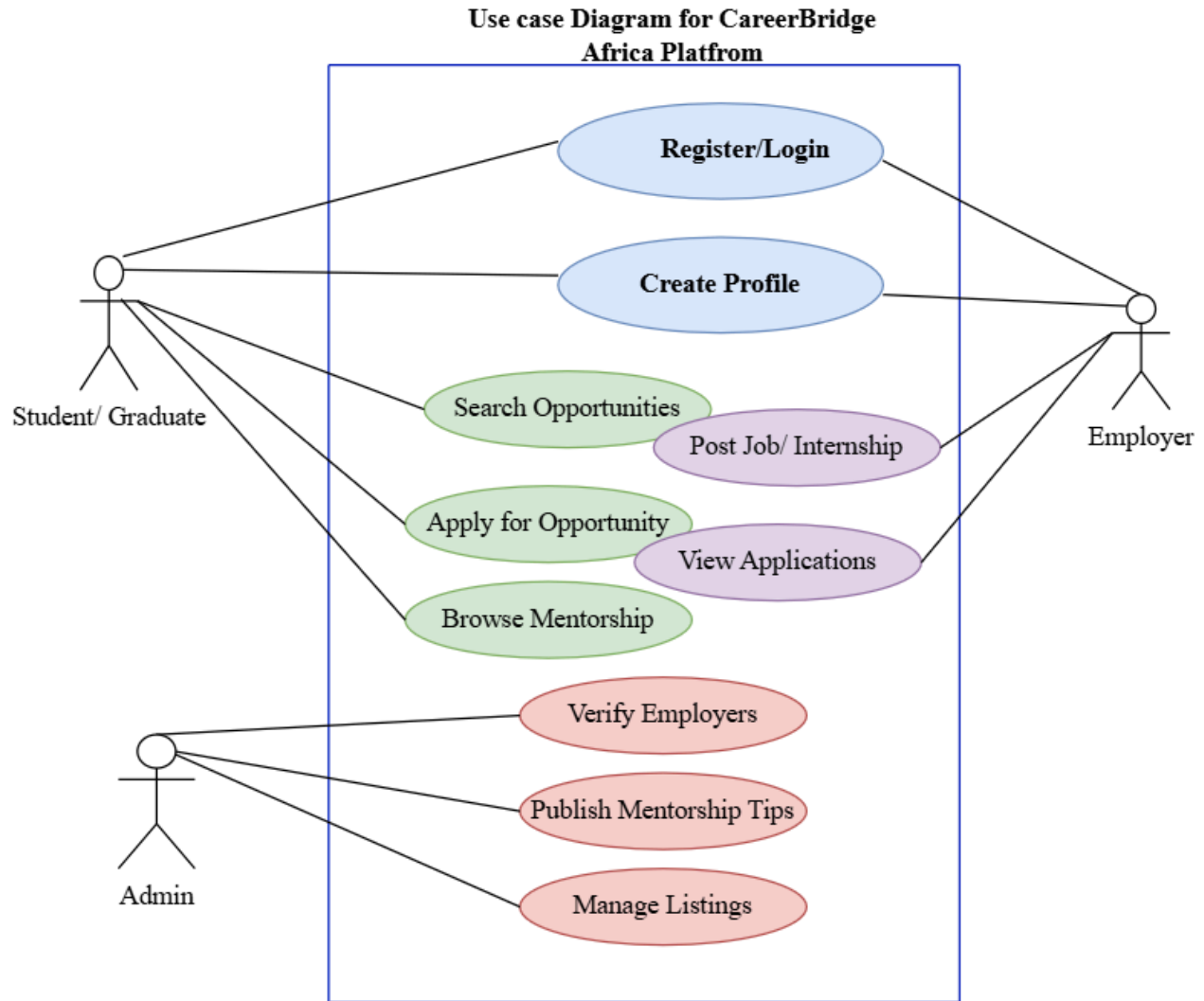
Appendix B: Analysis Models

Use case diagram

This Use Case Diagram shows how the users use the CareerBridge Africa platform and the major functionality of the system. There are three main actors identified: Student/Graduate, Employer, and Admin.

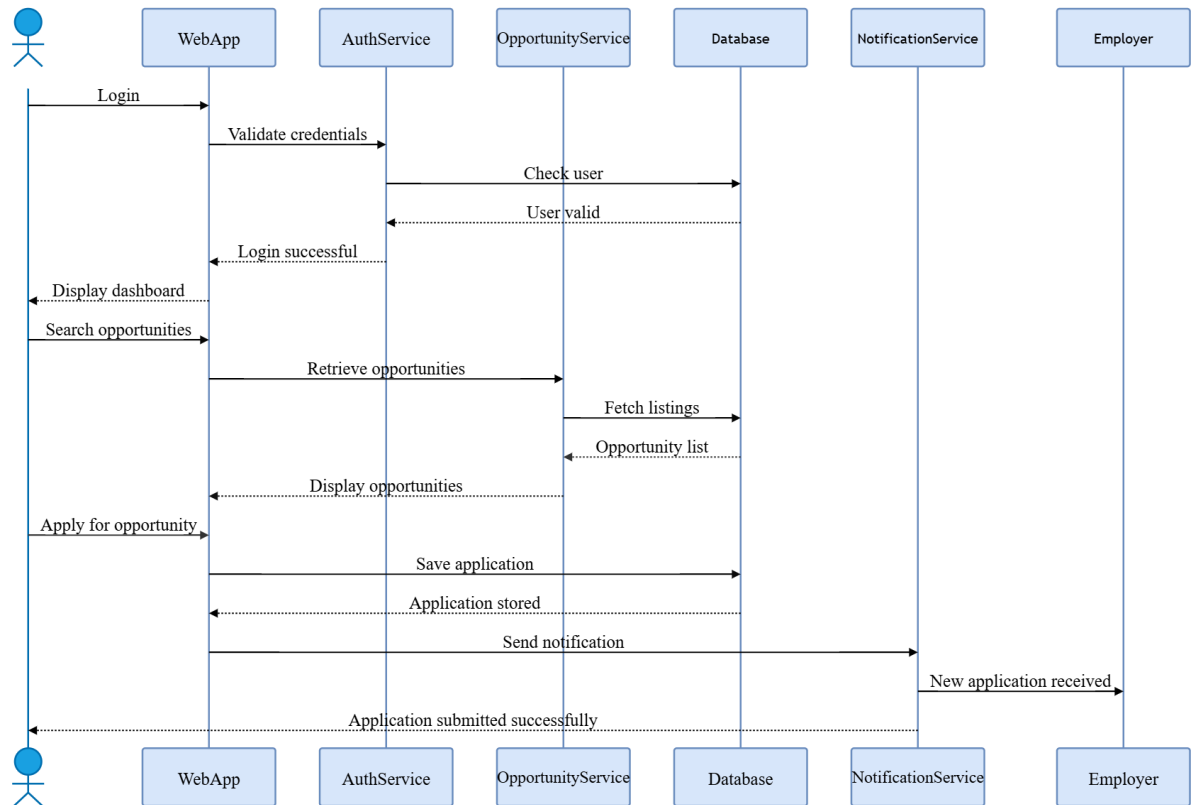
Students and graduates can register or sign in, create their profiles, search and apply for job or internship opportunities, and access mentoring materials. Employers can register on the platform, create an employer profile, publish job or internship offers, and evaluate applicants' applications. Admin performs verification of the employers, publishing of mentorship tips, and job listing management to maintain the security of the platform.

It is shown in the relations between actors and functions that are available to the particular actor, and it demonstrates how users interact with the system to meet its goal.



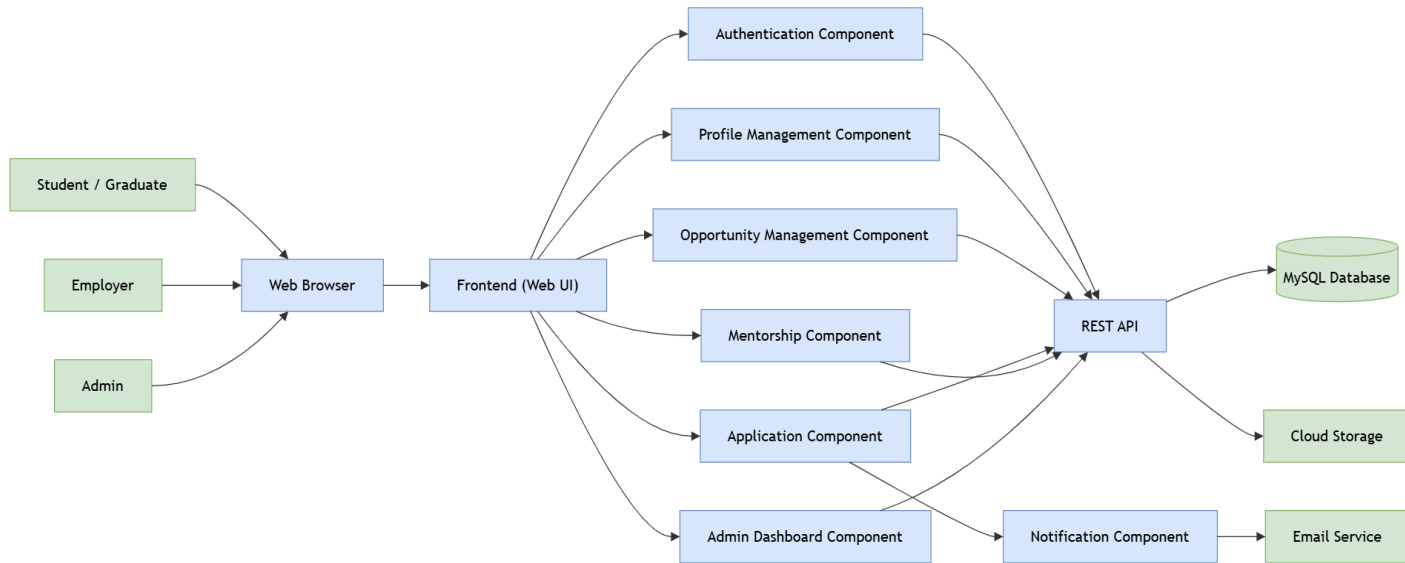
Sequence Diagram

This sequence diagram represents the process flow involved when a student applies for a job opportunity. Here, the student logs in, and the system uses the Authentication Service to verify the user's information. Then, the system looks for the opportunities using the service known as Opportunity Service from the database after successful login. At this point, the student applies for the opportunity, and the application is saved in the database. Finally, the Notification Service sends an alert to the employer regarding the new application.



Component Diagram

This Component Diagram describes the whole architecture of the CareerBridge Africa software. The users can get to the website using a web browser and then use the Frontend component of the site. The Frontend connects to the different components that handle authentication, profiles, opportunities, applications, mentorship, administration, and notifications. The different components then connect with the REST API, which has connections with the MySQL database and cloud storage. The Notification component sends emails from the Email Service when any important event happens, like successful applications or account notifications.



Class diagram

This Class Diagram shows the static view of CareerBridge Africa by showing the main classes, their attributes, operations, and relationships. In addition, the User class becomes the parent class of StudentGraduate, Employer, and Admin using inheritance since all these users have common information, including the user's name, email address, password, and user role. Students and graduates register in the system, search for and apply to opportunities, access mentorship resources, give feedback, and report fake jobs, while employers post opportunities. Administrators validate the identity of employers, manage job postings, and publish mentorship resources. Notifications get generated by applications to be delivered to users.

